

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience implementing AI/ML algorithms in Python at Deutsche Bundesbank (AnaCredit) to automate manual processes, paired with coursework and project experience in machine learning and AI. Comfortable programming AI-based solutions end-to-end and working closely with a development team to continuously improve software architecture.

Core Competencies

- **AI/ML Algorithm Implementation:** Python-based implementation of rule-based and ML detection algorithms (K-Means, Random Forest, SVM, Neural Networks) to automate manual workflows
- **Programming:** Python (Pandas, NumPy, Scikit-Learn), SQL; building and extending software components and data interfaces
- **Software Architecture Collaboration:** Experience working closely with a technical team to continuously improve automated tooling
- **Data Quality:** Automated deviation and anomaly detection, referential-integrity and consistency checks on operational data
- **Agentic AI Tools:** Regular use of Claude Code as an AI coding agent to accelerate development and automation
- **Collaboration & Documentation:** Stakeholder-facing requirement translation, Jira, Git, structured technical documentation

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Independently identified a recurring manual reporting process and digitalized it: built an automation pipeline generating Jira tickets and emails directly from detected data errors
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames, verifying and validating operational data before roll-out
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations) in reporting data, then produced clear analyses and KPI summaries of findings
 - Significantly reduced response time to data quality issues through end-to-end ownership, from detection to resolution

Projects

- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization *10/2024–12/2024*
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to segment high-risk groups, directly analogous to clustering structured records into meaningful categories

- **Personal Project (GitHub)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; built using **Claude Code** as an AI coding assistant to accelerate implementation
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026
- **Tecnológico de Monterrey** **Mexico**
Abitur 2019–2022
Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.